

HashSet & TreeSet — Coding Questions

20 Questions with Test Cases and Expected Outputs

Q1. Remove Duplicates

Given an array of integers, use HashSet to remove duplicate elements.

Test Case 1: Input: 10 20 10 30 20 40

Output: 10 20 30 40

Test Case 2: Input: 5 5 5 10 20 10

Output: 5 10 20

Q2. Count Unique Elements

Find the number of unique elements using HashSet.

Test Case 1: Input: 1 2 2 3 4 4 5

Output: 5

Test Case 2: Input: 10 10 20 20 30

Output: 3

Q3. Check Duplicate Elements

Use HashSet to determine whether an array contains duplicates.

Test Case 1: Input: 10 20 30 20 40

Output: Duplicate Found

Test Case 2: Input: 10 20 30 40

Output: No Duplicate

Q4. Find Common Elements

Given two arrays, find their common elements using HashSet.

Test Case 1: Input: A = 10 20 30 40; B = 20 40 50 60

Output: 20 40

Test Case 2: Input: A = 1 2 3; B = 4 5 6

Output: No Common Element

Q5. Find Union of Two Arrays

Find the union of two arrays using HashSet.

Test Case 1: Input: A = 1 2 3 4; B = 3 4 5 6

Output: 1 2 3 4 5 6

Test Case 2: Input: A = 10 20; B = 20 30

Output: 10 20 30

Q6. First Array but Not Second

Find elements present in the first array but not in the second.

Test Case 1: Input: A = 1 2 3 4 5; B = 2 4

Output: 1 3 5

Test Case 2: Input: A = 10 20 30; B = 10 30

Output: 20

Q7. Remove Duplicate Words

Given words, remove duplicate words using HashSet.

Test Case 1: Input: Java is easy Java is powerful Java

Output: Java is easy powerful

Test Case 2: Input: apple banana apple mango banana

Output: apple banana mango

Q8. Store Numbers in Sorted Order

Use TreeSet to store integers and display them in ascending order.

Test Case 1: Input: 50 10 40 20 30

Output: 10 20 30 40 50

Test Case 2: Input: 5 1 5 3 2 4

Output: 1 2 3 4 5

Q9. Smallest and Largest

Use TreeSet to find the smallest and largest element.

Test Case 1: Input: 45 12 89 23 7 56

Output: Smallest = 7, Largest = 89

Test Case 2: Input: 100 25 75 10

Output: Smallest = 10, Largest = 100

Q10. Elements Greater Than Given Number

Display all TreeSet elements greater than a given number.

Test Case 1: Input: 10 20 30 40 50 60; Given = 30

Output: 40 50 60

Test Case 2: Input: 5 15 25 35 45; Given = 20

Output: 25 35 45

Q11. Elements Less Than Given Number

Display all TreeSet elements less than a given number.

Test Case 1: Input: 10 20 30 40 50; Given = 35

Output: 10 20 30

Test Case 2: Input: 5 15 25 35; Given = 25

Output: 5 15

Q12. Elements Between Two Values

Display TreeSet elements within a specified range.

Test Case 1: Input: 10 20 30 40 50 60 70; Range = 25 to 60

Output: 30 40 50 60

Test Case 2: Input: 5 10 15 20 25 30; Range = 10 to 25

Output: 10 15 20 25

Q13. Next Greater Element

Find the next greater element using TreeSet higher().

Test Case 1: Input: 10 20 30 40 50; Given = 25

Output: 30

Test Case 2: Input: 5 15 25 35; Given = 15

Output: 25

Q14. Previous Smaller Element

Find the previous smaller element using TreeSet lower().

Test Case 1: Input: 10 20 30 40 50; Given = 25

Output: 20

Test Case 2: Input: 5 15 25 35; Given = 25

Output: 15

Q15. Remove Duplicates and Sort

Remove duplicates and display elements in sorted order using TreeSet.

Test Case 1: Input: 40 10 20 40 30 10 50

Output: 10 20 30 40 50

Test Case 2: Input: 8 3 8 1 5 3 2

Output: 1 2 3 5 8

Q16. Second Largest Element

Use TreeSet to find the second largest distinct element.

Test Case 1: Input: 10 50 20 80 30 80 40

Output: Second Largest = 50

Test Case 2: Input: 5 10 10 20 30

Output: Second Largest = 20

Q17. Second Smallest Element

Use TreeSet to find the second smallest distinct element.

Test Case 1: Input: 30 10 50 20 10 40

Output: Second Smallest = 20

Test Case 2: Input: 5 5 10 15 20

Output: Second Smallest = 10

Q18. Find Missing Number

Numbers from 1 to n are given with one number missing. Use HashSet to find it.

Test Case 1: Input: 1 2 3 4 5 6 8 9 10

Output: Missing Number = 7

Test Case 2: Input: 1 2 3 5 6

Output: Missing Number = 4

Q19. Find Duplicate Elements

Use HashSet to find all elements that occur more than once.

Test Case 1: Input: 1 2 3 2 4 5 1 6 3

Output: 1 2 3

Test Case 2: Input: 10 20 10 30 20 20

Output: 10 20

Q20. Student Names in Sorted Order

Store student names in TreeSet. Remove duplicates and display alphabetically.

Test Case 1: Input: Rahul, Aman, Priya, Rahul, Karan, Aman

Output: Aman, Karan, Priya, Rahul

Test Case 2: Input: Neha, Amit, Neha, Riya, Amit

Output: Amit, Neha, Riya